Due on tuesday, March 21st, in class

Long, for the Study Break. Please start work early.

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1. A subatomic particle lives for 6×10^{-7} seconds, i.e., it decays 6×10^{-7} seconds after it is created, as seen from its own reference frame. This particle is created at a speed $v = \frac{4}{5}c$ with respect to earth,

   (a) **[3 pts.]** How long does it live, according to the earth frame? Specify clearly: in which frame is the lifetime the 'proper' time?

   (b) **[2 pts.]** How far will the particle travel before it decays, as seen from the earth frame?

   (c) **[2 pts.]** Imagine that the Galilean transformations were true instead of the relativistic (Lorentz) transformations. If this were the case, how far would the particle travel before it decays, as seen from the earth frame?

2. Frame S' moves at speed v_1 with respect to frame S, and frame S'' moves at speed v_2 with respect to frame S', both in the common x, x', x'' direction. The Lorentz transformations are

$$x' = \gamma_1(x - v_1 t)\ , \quad t' = \gamma_1(t - v_1 x/c^2) \qquad \text{with} \quad \gamma_1 = \gamma(v_1) = \frac{1}{\sqrt{1 - v_1^2/c^2}}$$

$$x'' = \gamma_2(x' - v_2 t')\ , \quad t'' = \gamma_2(t' - v_2 x'/c^2) \qquad \text{with} \quad \gamma_2 = \gamma(v_2) = \frac{1}{\sqrt{1 - v_2^2/c^2}}$$

   (a) **[3 pts.]** Express x'' (the coordinate of the S'' frame) in terms of x and t, in the form $x'' = Ax + Bt$. Express A and B in terms of v_1, v_2, γ_1 and γ_2.

   (b) **[2 pts.]** If frame S'' moves at speed u with respect to frame S, you would have $x'' = \gamma(u)(x - ut)$. Comparing with $x'' = Ax + Bt$ and using your expressions for A and B, find u. Have you recovered the relativistic velocity addition formula?

   (c) **[2 pts.]** Verify that your constant A is really equal to $\gamma(u)$, with the u known from the velocity addition formula.

3. Frame Σ' moves at speed v with respect to frame Σ, in the common x, x' direction. The two frames are aligned at time $t = t' = 0$.

   (a) **[4+2 pts.]** A particle has velocity $\vec{u}' = (0, c, 0)$ relative to Σ'. Use the velocity addition formulae to find the velocity of the particle relative to Σ.

   Explain how your result is consistent with the constancy of the speed of light.

   (b) **[2 pts.]** A stick is at rest in frame Σ. It lies in the xy plane and makes angle $\pi/4$ with the x-axis. Using the Lorentz transformations or the length contraction formula, find the angle that the stick makes with the x' axis, as observed from the Σ' frame.

4. A stick of proper length L moves past you at speed v. There is a time interval between the front end passing you (first event) and the back end passing you (second event). We are interested in this time interval and in the length of the stick, as seen from various frames.

   (a) **[1 pt.]** Draw snapshots of the two events as seen from your frame, marking with an arrow what is moving (you or the stick).

   (b) **[4 pts.]** Find the length of the stick as seen from your frame, and the time between the two events, as seen from your frame.

   (c) **[1 pts]** Draw snapshots of the two events as seen from the frame of the stick, marking with an arrow what is moving, as seen from this frame.

   (d) **[4 pts.]** Find the length of the stick and the time between the two events, as seen from the frame of the stick.

   (e) **[2 pts.]** In which frame is the time interval the *proper* time interval: your frame or the stick's frame? Explain why.

5. Frame Σ' moves at speed v with respect to frame Σ, in the common x, x' directions. The two frames are aligned at time $t = t' = 0$.

   (a) **[2 pts.]** Write down the Galilean transformation as a matrix transforming the Σ-frame coordinates (ct, x, y, z) to the Σ' coordinates (ct', x', y', z').

   (b) **[2 pts.]** Write down the Lorentz transformation matrix and show how it reduces to the Galiliean transformation matrix in the limit $c \to \infty$.

   (c) **[3 pts.]** Considering two successive Galilean transformations in the same x direction, with velocities v_1 and v_2, show by matrix multiplication that their combination is also a Galilean transformation, with velocity $v_1 + v_2$.

6. Some groups. This problem requires some (serious) searching and reading.

   Look up the terms $SO(2)$, $SO(3)$, $SO(3,1)$, $SU(2)$. Each of these groups can be thought of as a set of matrices.

   (a) **[2 pts.]** What does 'S' in these names stand for?

   Which matrices are included in the group $O(3)$, but not in the group $SO(3)$?

   (b) **[2 pts.]** Which physical transformations are represented by the groups $SO(2)$ and $SO(3)$? Are these groups abelian or non-abelian?

   (c) **[1 pt.]** What is the physical interpretation of matrices which are included in the group $O(3)$, but not in the group $SO(3)$?

   (d) **[1 pt.]** How many matrices does $SO(2)$ contain?

   (e) **[3 pts.]** The group $SO(3,1)$ is the proper Lorentz group.

   What is the size of matrices belonging to this group?

   These matrices are not orthogonal, but must instead satisfy a similar equation. Write down the equation that defines membership in the Lorentz group.

   Under what condition is a member of this group called *orthochronous*?